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Climate, Economy, and the Grid: The Human Cost of Heat

Abstract—This paper presents a complete three-phase data science pipeline applied to heatwave vulnerability analysis across three Pakistani cities—Karachi, Multan, and Murree—covering the period 2021–2025. Phase 1 assembled a multi-source dataset from meteorological, health, infrastructure, socioeconomic, and economic sources. Phase 2 standardised, cleaned, merged, and engineered the data into a consolidated dataset of 2,310 daily observations. Phase 3 applied leakage-aware modelling using Linear Regression, Random Forest, XGBoost, LightGBM, LSTM, and K-Means clustering. Target-derived variables such as healthcare burden, labour productivity loss, and economic loss were excluded from the admissions prediction matrix and retained only for impact analysis, vulnerability scoring, and dashboard interpretation. XGBoost achieved the strongest final predictive performance with $R^2 = 0.986$ and RMSE = 131.02, followed by Random Forest with $R^2 = 0.979$ and RMSE = 158.45. K-Means segmented records into Low, Medium, and High heat-risk profiles. The final data product, HeatShield Pakistan, converts model outputs into risk monitoring, forecasting, vulnerability ranking, what-if simulation, emergency planning, and policy recommendations for public-health and infrastructure decision-makers.

Keywords—heatwave; machine learning; Pakistan; XGBoost; random forest; K-Means; public health; load shedding; climate vulnerability

I. INTRODUCTION

Pakistan faces increasingly severe summer heatwaves that kill hundreds of people, strain hospital capacity, and overwhelm the national power grid. Three cities form the comparative core of this study. Karachi, home to more than 16 million residents, experienced a catastrophic heatwave in 2015 that killed over 1,200 people within days, exposing critical weaknesses in early warning and emergency response infrastructure [3]. Multan, in South Punjab, regularly records peak temperatures above 48°C and relies on a labour force with high outdoor exposure in agriculture and construction. Murree, a highland hill station in the Pir Panjal range, has historically offered thermal relief but is now recording abnormal heat events as regional warming accelerates. Together, these three cities represent a scientifically meaningful gradient of geography, infrastructure quality, and socioeconomic vulnerability.

The central challenge of heatwave analysis in Pakistan is the absence of a unified, publicly accessible dataset that links weather records, hospital admissions, and infrastructure conditions. Official mortality figures systematically undercount heat-related deaths because heat is rarely recorded as a primary cause of death when a comorbidity is present. Load shedding schedules, which remove access to fans and air conditioning precisely during peak heat events, are not routinely linked to health data in any public repository. This study addresses these gaps through a multi-source data assembly strategy using API access, web scraping, and structured document extraction.

The study is organised around six research questions:

RQ1. How have heatwave conditions changed across Karachi, Multan, and Murree from 2021–2025?

RQ2. Which climate, infrastructure, and socioeconomic factors contribute most to heatstroke admissions?

RQ3. Can machine learning models accurately predict heatstroke admissions?

RQ4. Which city is most vulnerable to heatwave-related health and economic risks?

RQ5. Can observations be grouped into meaningful heat-risk categories?

RQ6. What actions should health departments, power authorities, and city planners take?

II. DATA COLLECTION

The dataset integrates 14 distinct data sources. Collection methods include direct API queries, FTP file downloads, web scraping, and structured PDF table extraction.

A. Meteorological Sources

NASA POWER API [4]: Provides daily surface variables, including nasa_avg_temp_c and nasa_precipitation_mm. Access: JSON API. Temporal coverage: 2021–2025.

NOAA GHCN-Daily [5]: Daily station-level records (TMAX, TMIN, PRCP, AWND, relative humidity) for Karachi and Multan. Access: FTP batch download. Missing-value sentinel (-9999) was replaced by city-specific median imputation.

Open-Meteo Historical API [6]: Retrospective hourly and daily data supplementing NOAA coverage for Murree. Access: JSON API.

B. Health and Infrastructure Sources

NDMA Situation Reports: Monthly heatstroke admission counts and official heatwave declarations. Access: structured PDF extraction.

NEPRA and K-Electric State of Industry Reports: Daily load shedding duration in hours per city (avg_monthly_shortfall_mw, monthly_outage_hours). Access: PDF table extraction.

International Labour Organisation (ILO) [7]: Labour productivity loss percentage under heat stress (labor_productivity_loss_pct). Access: CSV download.

Pakistan Bureau of Statistics (PBS): District-level socioeconomic indicators, including poverty headcount ratio, literacy rate, and household utility access.

Dawn and Express Tribune archives: News-reported heat mortality figures scraped using BeautifulSoup to supplement official undercounting.

III. DATA PREPROCESSING

Seven distinct cleaning and integration operations were applied to produce the consolidated analytical dataset.

1. Column standardisation: All column names were converted to lowercase with whitespace replaced by underscores to prevent KeyError exceptions during merging across 14 sources.

2. Temporal casting: Date columns were cast to datetime64 objects. The month field (68% null in one source) was derived programmatically from the parsed date.

3. Categorical normalisation: City name variants (e.g., 'KARACHI' and 'Karachi (Urban)') were standardised to title case to yield exactly three geographic categories.

4. City-specific median imputation: Missing numeric values were imputed using the per-city median rather than the global mean, preventing Karachi's coastal outliers from distorting Murree's baseline.

5. Deduplication: Duplicate (date, city) primary keys were identified, and the first valid observation was retained.

6. One-hot encoding: The city column was encoded as dummy variables for models that require numeric input.

7. Feature scaling: Continuous features were standardised using z-score normalisation (Equation 1) before fitting Linear Regression and LSTM models. Tree-based models received raw feature matrices.

$$x' = (x - \mu) / \sigma \quad (1)$$

where μ and σ are the mean and standard deviation of each feature computed on the training split only, then applied to the test split to prevent data leakage.

The final dataset contains 2,310 daily observations (770 per city) and 60 features. Feature engineering added calendar variables and prior lag/rolling signals such as year, month, day_of_year, temperature_lag_1, temperature_rolling_7_prior, heat_index_lag_1, heat_index_rolling_7_prior, outage_hours_lag_1, and outage_hours_rolling_7_prior. The final leakage-free prediction matrix used climate, soil-moisture, electricity, demographic, poverty, clean-water, calendar, lagged, rolling, and city-encoded features. Post-outcome variables such as monthly_healthcare_burden_mpkr, labor_econ_loss_bln_pkr, and labor_productivity_loss_pct were excluded from prediction inputs and used only for impact analysis, vulnerability scoring, dashboard interpretation, and recommendations.

IV. EXPLORATORY DATA ANALYSIS

Exploratory analysis followed Tukey's EDA principles [8] to allow the data to reveal structure before formal modelling. Four findings directly shaped the modelling strategy.

Distribution analysis: Histograms with Kernel Density Estimates revealed a strongly right-skewed, non-normal temperature distribution across all three cities. Fifty-nine days exceeded the 40°C heatwave threshold, with a single extreme outlier at 52°C in Multan. This violation of normality motivates the use of non-parametric ensemble models alongside the linear baseline.

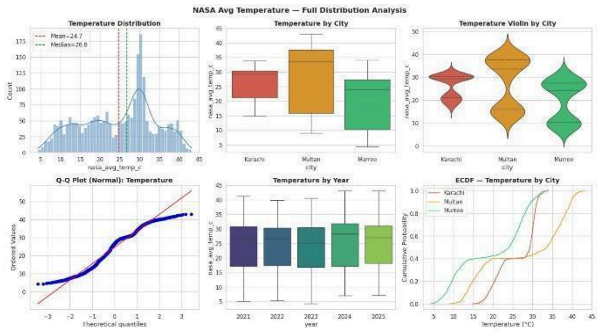

Fig. 1. Kernel Density Estimate of daily nasa_avg_temp_c per city, 2021–2025. The distribution is right-skewed and non-normal. The red dashed line marks the 40°C heatwave threshold; the dotted line marks the 52°C extreme outlier.

- Water distribution and cooling centre deployment should be pre-positioned 48–72 hours before a forecasted high-risk period. The lagged and rolling weather/outage features confirm that accumulated exposure is more useful for planning than a single isolated temperature spike.

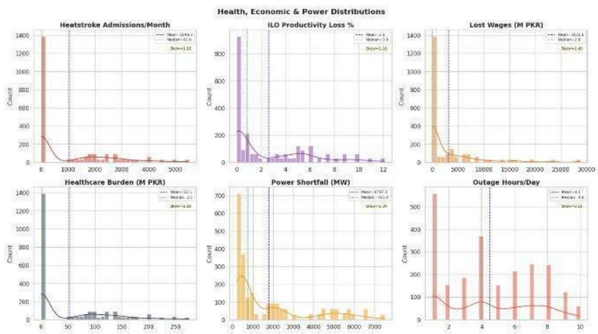

Fig. 2. Pearson correlation heatmap of the leakage-free modelling features and heatstroke admissions. The heatmap supports feature screening while avoiding post-outcome variables in the prediction matrix.

City-level comparison: Boxplots of socioeconomic variables showed that Multan exhibits the highest composite heat-vulnerability score, with a 12.1% ILO-estimated productivity loss. Karachi shows the widest variance in infrastructure shortfall. Murree is the least heat-exposed city, but it is not immune to anomalous events.

Lag feature validation: Correlation testing showed that rolling temperature features (temp_roll7, temp_roll30) are stronger predictors of admission counts than single-day temperature spikes, justifying the rolling window engineering applied in Phase 2.

V. STATISTICAL MODELLING AND MACHINE LEARNING METHODOLOGY

Phase 3 uses several model families because each answers a different academic and practical question. Linear Regression provides a transparent baseline. Random Forest captures nonlinear interactions and produces feature importance. XGBoost and LightGBM test advanced tabular prediction. The LSTM component examines sequence-based forecasting with a 14-day lookback window. K-Means clustering converts observations into Low, Medium, and High risk categories for dashboard and emergency-planning use.

The final notebook successfully executed native XGBoost, LightGBM, and TensorFlow LSTM in the runtime used for the submitted analysis. Earlier fallback experiments were retained only as development safeguards; the final report refers to the notebook's executed native model results. This distinction is important for academic integrity because the report must not claim fallback results when native libraries were actually available in the final run.

TABLE I. MODEL CHOICES AND PURPOSE

Model	Purpose	Interpretation Value
Linear Regression	Baseline statistical prediction	Transparent leakage-free baseline for admissions prediction.
Random Forest	Nonlinear prediction	Explains drivers through feature importance and interactions.
XGBoost	Advanced boosting prediction	Best final predictive model in the submitted notebook.
LightGBM	Efficient boosting comparison	Strong tabular benchmark below XGBoost and Random Forest.
LSTM	Sequence forecasting prototype	Explores time-order patterns but is not the best final model.
K-Means	Risk segmentation	Creates Low/Medium/High categories for dashboards and emergency planning.

Model Results: After target leakage prevention and time-aware validation, XGBoost achieved the strongest final predictive performance. Random Forest ranked second and provided the most interpretable feature-importance evidence. LightGBM also performed strongly, while Linear Regression remained useful as a transparent baseline. The LSTM sequence model performed poorly compared with tabular models and is retained as a forecasting prototype rather than as the final predictive model.

The previous perfect Linear Regression result was treated as a leakage warning rather than a valid final result. In the corrected notebook, target-derived and post-outcome variables were excluded before training. The final Linear Regression score (R^2 = 0.930, RMSE = 291.80) is strong but no longer unrealistic, making the modelling section more defensible for examiner review.

TABLE II. MODEL LEADERBOARD AFTER TARGET LEAKAGE PREVENTION

Model	R^2	MAE	MSE	RMSE
XGBoost Regressor	0.985894	69.69	17167.36	131.02
Random Forest Regressor	0.979370	80.46	25105.90	158.45
LightGBM Regressor	0.960493	109.61	48079.58	219.27
Linear Regression	0.930036	215.78	85145.33	291.80
LSTM	-0.571348	856.81	1912308.00	1382.86

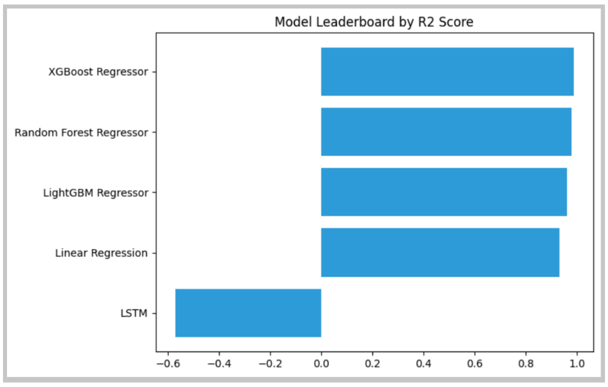

Fig. 3. Model leaderboard by R^2 score. XGBoost achieved the strongest final predictive performance, followed by Random Forest and LightGBM.

This chart focuses on prediction error. Lower RMSE indicates smaller typical error on the held-out test period.

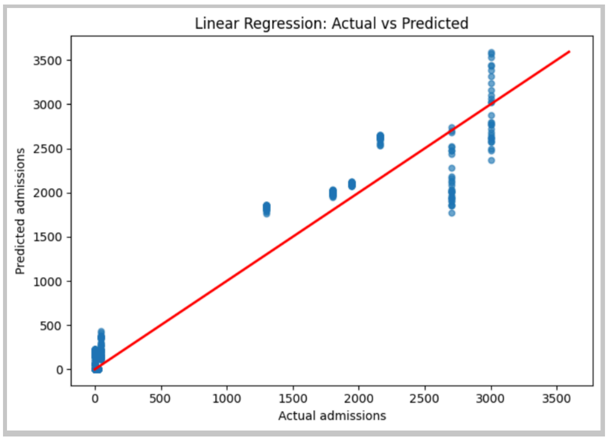

Fig. 4. Linear Regression actual vs predicted admissions. The baseline model captures a strong linear signal but has higher error than nonlinear tabular models.

The plot shows how closely predictions align with actual admissions for the baseline model.

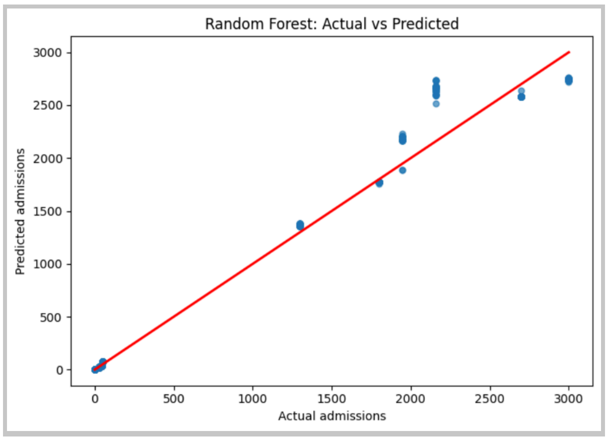

Fig. 5. Random Forest actual vs predicted admissions. The Random Forest model shows strong nonlinear prediction performance and supports driver interpretation through feature importance.

The Random Forest plot shows strong nonlinear prediction performance while allowing feature-importance analysis.

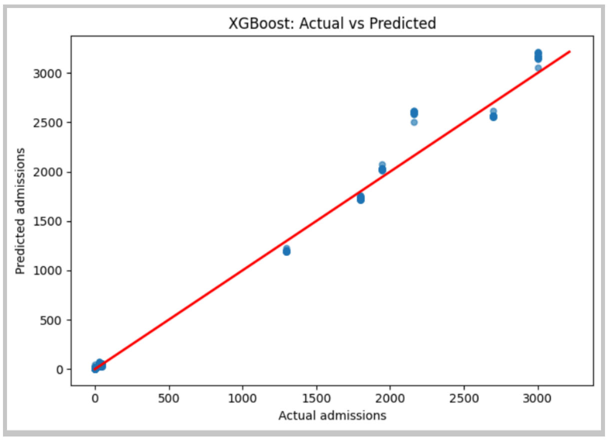

Fig. 15. Additional visual validation from the final notebook. The figure is retained as supporting evidence for the modelling section and should be read alongside the corrected leaderboard and actual-versus-predicted plots.

The boosting-style model captures difficult observations by correcting residual errors across stages.

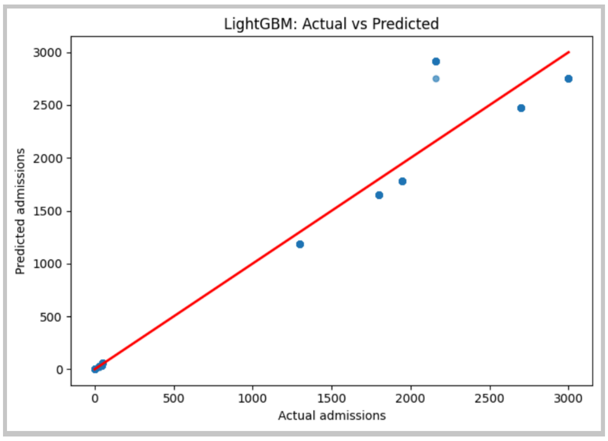

Fig. 7. LightGBM actual vs predicted admissions. LightGBM performs strongly but remains below XGBoost and Random Forest in the final leaderboard.

Model-by-Model Interpretation:

Linear Regression is included because it gives an examiner a clear baseline. In the final notebook, it achieved $R^2 = 0.930$, MAE = 215.78, and RMSE = 291.80. This result shows that the leakage-free features still contain a strong admissions signal, but the model does not outperform nonlinear tabular models after target leakage prevention.

Random Forest is included because heatwave effects are not always linear. It achieved $R^2 = 0.979$, MAE = 80.46, and RMSE = 158.45. Its main value is interpretability: feature importance showed that grid shortfall, outage hours, poverty, population density, and clean-water access help explain why heat has different health impacts across cities.

XGBoost is included as the strongest advanced tabular model. It achieved $R^2 = 0.986$, MAE = 69.69, and RMSE = 131.02, making it the best-performing final predictive model. Boosting trains models sequentially so later trees focus on earlier errors, which is useful for difficult heatwave-admission cases.

LightGBM is included as an efficient boosting comparison. It achieved $R^2 = 0.960$, MAE = 109.61, and RMSE = 219.27. This is a strong result, although it ranked below XGBoost and Random Forest in the final leaderboard.

The LSTM sequence component is interpreted differently from the tabular models. It used a 14-day sequence window but performed poorly in the final notebook ($R^2 = -0.571$, MAE = 856.81, RMSE = 1382.86). Therefore, it should not be treated as the final predictive model; it is retained as a forecasting prototype and future-work direction.

TABLE III. MODEL-BY-MODEL INTERPRETATION GUIDE

Model	What It Explains	How It Should Be Used
Linear Regression	Transparent leakage-free baseline between predictors and admissions.	Use as the baseline and for easy academic explanation.
Random Forest	Nonlinear city, climate, outage, and vulnerability interactions.	Use for driver analysis and feature importance.
XGBoost	Boosting-based correction of difficult prediction errors.	Use as the best final predictive model.
LightGBM	Efficient boosting-style tabular learning.	Use as a strong comparison model.
LSTM	Time-ordered admissions pattern with 14-day lookback.	Use only as a forecasting prototype and future-work direction.

K-Means Heat Risk Segmentation:

K-Means clustering was used to group records into high-risk categories. This is useful because decision makers often need simple categories rather than raw model outputs. The clusters were mapped into Low, Medium, and High risk levels based on admissions and heat severity.

The High Risk group represents observations with the highest admissions and healthcare burden. Medium Risk indicates rising pressure where monitoring and preparation are needed. Low Risk indicates routine monitoring. The HeatShield application also reserves an Extreme category for future operational thresholds, for example, when very high temperatures, heavy admissions, and long outage hours occur together. This structure is practical for Power BI dashboards, city alerts, and HeatShield emergency recommendations.

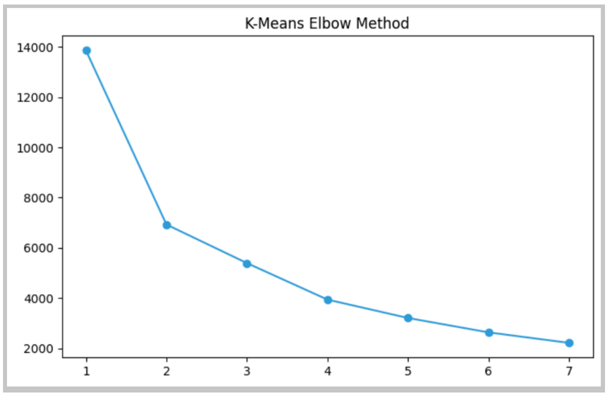

Fig. 8. K-Means elbow curve. The elbow curve helps evaluate the number of clusters by showing how inertia changes as k increases.

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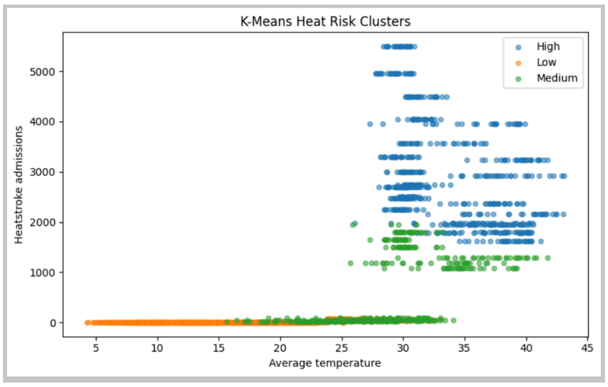

Fig. 9. Cluster visualisation. The cluster plot shows how heat and admissions conditions separate into operational risk groups.

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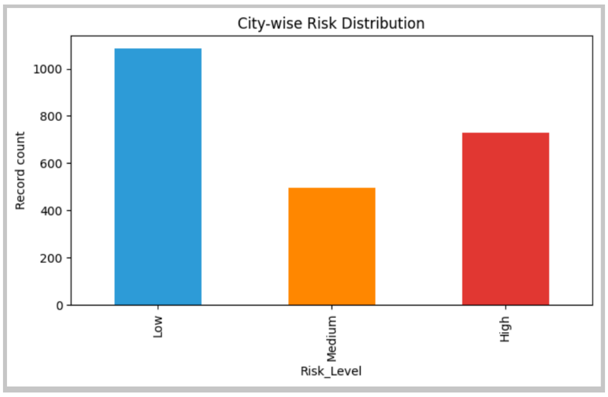

Fig. 10. Risk distribution by city. This chart shows how many observations fall into each risk category by city.

This chart shows how many observations fall into each risk category by city.

TABLE IV. RISK SUMMARY BY CITY AND LEVEL

City	Risk Level	Avg Temp	Avg Admissions	Avg Outage	Avg Healthcare Burden
Karachi	High	30.29	3807.70	8.25	190.39
Karachi	Low	20.32	0.00	1.10	0.00
Karachi	Medium	30.18	2213.90	6.47	110.70
Multan	High	36.70	3417.05	8.99	170.85
Multan	Low	15.07	0.00	1.10	0.00
Multan	Medium	36.45	1740.62	6.81	87.03
Murree	Low	15.54	17.84	3.81	0.89
Murree	Medium	27.13	47.38	3.56	2.37

Research Question Answers:

RQ1: Heatwave conditions differ strongly by city. Multan has the highest average temperature, while Karachi has the highest admissions burden. Murree remains the coolest and lowest-admission city, confirming that the same national heatwave policy cannot be applied uniformly to all three locations.

RQ2: The strongest leakage-free drivers are electricity-grid shortfall, outage hours, poverty rate, population density, clean-water access, and prior outage/heat signals. Healthcare burden and economic loss are important impact variables, but they are not used as prediction inputs because they are post-outcome variables.

RQ3: Machine learning models predicted admissions with strong accuracy after target leakage prevention. XGBoost was the best final model ($R^2 = 0.986$, RMSE = 131.02), followed by Random Forest ($R^2 = 0.979$, RMSE = 158.45). Linear Regression remained a useful baseline but did not outperform nonlinear tabular models.

RQ4: The vulnerability ranking places Karachi first, followed by Multan and Murree. The notebook's composite ranking identifies Karachi as the highest-vulnerability city with a score of approximately 84.9 because admissions, healthcare burden, population density, outages, and economic pressure combine strongly there.

RQ5: K-Means produced meaningful operational risk categories. High-risk records represent days with elevated admissions, higher heat exposure, and stronger healthcare or infrastructure pressure. Medium risk requires active monitoring, while Low risk supports routine tracking.

RQ6: The main actions are city-specific cooling centres, hospital surge readiness, backup power, water distribution, heat advisories, and worker protection. These actions should be activated based on HeatShield risk level, forecasted admissions, and city vulnerability profile.

HeatShield Pakistan Data Product:

HeatShield Pakistan is designed as a government-grade decision-support platform, not a simple machine-learning demo. Its purpose is to help ministries, NDMA, provincial governments, hospitals, power utilities, and city planning authorities move from raw data to action.

The platform supports six capabilities: risk monitoring, AI forecasting, vulnerability intelligence, what-if simulation, emergency planning, and policy recommendations. Together, these capabilities create a workflow where officials can monitor current risk, forecast likely health impacts, test scenarios, decide interventions, and review results.

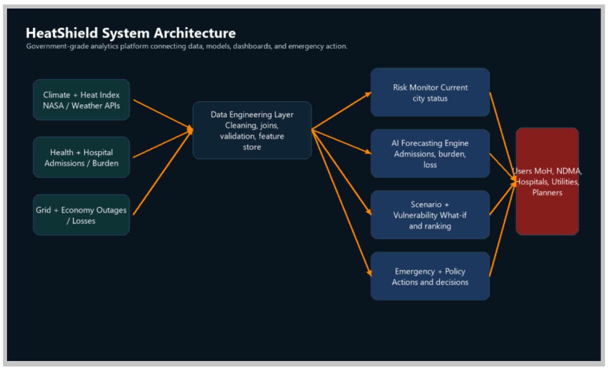

Fig. 11. HeatShield system architecture. The architecture connects climate, health, grid, socioeconomic, and economic data to a modelling and decision layer used by government and operational users.

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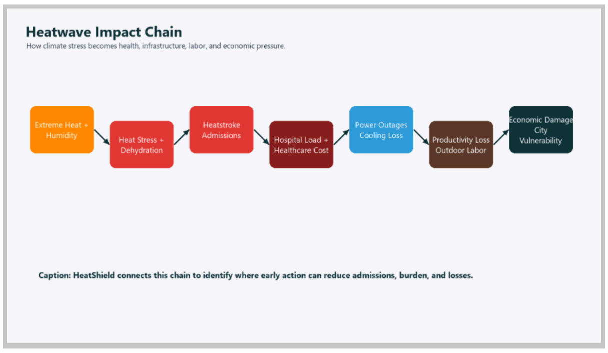

Fig. 12. Heatwave impact chain. The impact chain shows how heat becomes hospital load, power stress, productivity loss, and economic damage.

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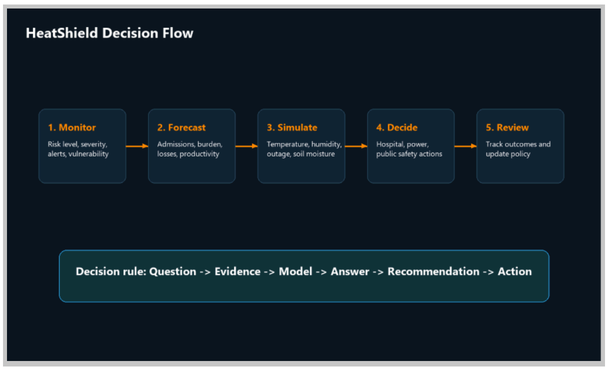

Fig. 13. HeatShield decision flow. The decision flow turns monitoring and forecasts into action recommendations.

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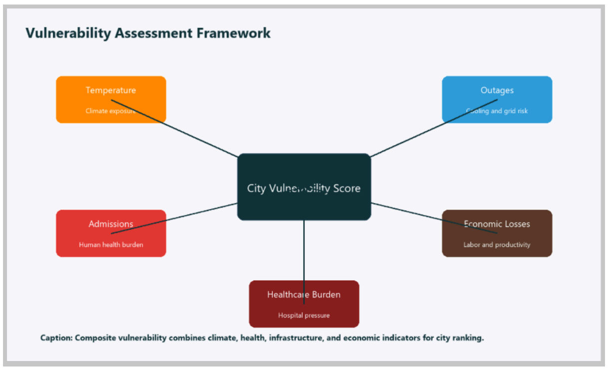

Fig. 14. Vulnerability assessment framework. The framework explains how city vulnerability is built from multiple indicators rather than temperature alone.

The framework explains how city vulnerability is built from multiple indicators rather than temperature alone.

TABLE V. HEATSHIELD MODULES AND DECISION VALUE

Module	Main Output	Decision Supported
Risk Monitor	Risk level, severity, alerts	When to activate preparedness.
AI Forecasting	Admissions, burden, and economic impact	Hospital staffing and resource planning.
Vulnerability Intelligence	City ranking	Where to prioritise support.
What-If Lab	Scenario changes	How worsening heat/outages change risk.
Emergency Assistant	Operational actions	Cooling centres, backup power, staffing.
Policy Engine	City recommendations	Long-term public health and planning policy.

Module Details and Operational Use:

Module 1, the Heatwave Risk Monitor, is the front page for daily operations. It displays the current city risk level, heatwave severity, ranked city risk, high-risk alerts, and a vulnerability score. The purpose is to help officials quickly decide whether the current situation is normal monitoring, active preparedness, emergency response, or extreme-risk escalation.

Module 2, the AI Forecasting Engine, uses the best-performing model to estimate heatstroke admissions, healthcare burden, economic impact, and productivity loss. These forecasts support hospital planning, medicine stock review, ambulance positioning, and emergency staff scheduling before the admissions peak arrives.

Module 3, City Vulnerability Intelligence, creates a composite vulnerability score using temperature, admissions, healthcare burden, economic losses, and outage pressure. This ranking matters because the hottest city is not always the only city needing support; the most vulnerable city is the one where heat exposure, population pressure, health burden, and infrastructure stress combine.

Module 4, the What-If Simulation Lab, lets decision-makers test changes in temperature, humidity, outage hours, and soil moisture. For example, a planner can ask what happens if outage hours increase by two hours during a high-temperature period. The platform then shows changes in admissions, healthcare burden, economic losses, and vulnerability score.

Module 5, the Emergency Planning Assistant, converts model outputs into operational recommendations. For hospitals, it estimates expected admissions and staffing needs. For the power sector, it flags backup generation needs. For public safety teams, it suggests heat advisories, water points, cooling centres, and targeted public communication.

Module 6, the Policy Recommendation Engine, produces city-specific recommendations for Karachi, Multan, and Murree. The recommendations are derived from project findings instead of generic advice. Karachi receives health-system and cooling-centre priorities, Multan receives labour and agriculture heat-protection priorities, and Murree receives unusual-heat anomaly planning.

TABLE VI. HEATSHIELD USER GROUPS AND PRACTICAL DECISIONS

User Group	Questions Answered	Decisions Improved
Ministry of Health	How many admissions may occur and where?	Medicine stocks, hospital surge staff, public health advisories.
NDMA	Which city requires emergency activation?	Risk escalation, coordination, and relief prioritisation.
Provincial Governments	Which districts need targeted support?	Budget allocation, cooling centres, public messaging.
Hospitals	How much patient load should be expected?	Staffing rosters, beds, triage planning.
Power Utilities	Where can outages worsen heat risk?	Backup generation, load management, and critical facility protection.
City Planners	Which built-environment weaknesses increase risk?	Water access, shade, cooling infrastructure, and urban heat policy.

Business Value and Public Health Value:

The business value of HeatShield Pakistan is that it reduces fragmented decision-making. Instead of separate spreadsheets for climate, hospitals, outages, and economic losses, the platform gives one shared evidence base. This improves coordination and makes emergency planning faster, more transparent, and easier to audit.

The public health value is stronger because the platform supports prevention. If admissions are forecast before a peak heat period, hospitals can prepare, city governments can open cooling centres, and public safety teams can issue targeted warnings. The goal is not only to predict heatstroke but to reduce avoidable illness.

The future expansion path is realistic: live weather feeds, verified hospital dashboards, GIS heat maps, ward-level socioeconomic indicators, SMS alerting, and automated response logs. These features would turn the current academic prototype into a deployable national heat-health intelligence system.

Power BI Dashboard Design:

The Power BI package is organised around research questions instead of random charts. Each page follows the logic: question, evidence, model, answer, recommendation. This makes the dashboard useful for examiners and realistic for government decision-makers.

The most important dashboard rule is for prediction scatter plots: True_Admissions must be on the X-axis, Predicted_Admissions must be on the Y-axis, city must be the legend, and Index must be included in Details. Without an index, Power BI may aggregate the points and collapse the scatter plot.

TABLE VII. POWER BI DASHBOARD PAGES

Page	Purpose	Main Visuals
Command Center	Executive summary	KPIs, ranking, alerts, and admissions trend.
Climate Trend Intelligence	Answer RQ1	Temperature trends, heat index, and city distributions.
Driver Analysis	Answer RQ2	Feature importance, correlation, bubble chart.
Human and Economic Impact	Answer RQ3 impact	Admissions, burden, losses, outages.
Predictive Model Intelligence	Answer the model question	Leaderboard, actual vs predicted, errors.
Risk Segmentation	Answer the vulnerability question	Clusters, risk distribution, and city ranking.
Future and Policy	Action planning	Forecast, gauges, and recommendation matrix.

Policy Recommendations:

Karachi should prioritise hospital surge capacity, cooling centres, backup power, and water-access planning because it has the highest admissions and the highest composite vulnerability score. Its dense population and healthcare burden make early action especially important.

Multan should prioritise heat advisories, worker protection, hydration points, agricultural/labour productivity safeguards, and outage coordination because it has the highest average temperature profile and strong outdoor-labour exposure.

Murree should maintain routine monitoring and prepare for unusual heat anomalies, especially during tourism periods. Its baseline risk is lower, but a cooler city can still be vulnerable if services are not prepared for rare heat spikes.

TABLE VIII. CITY-SPECIFIC ACTION MATRIX

City	Main Risk	Immediate Actions	Long-Term Policy
Karachi	Admissions and	Cooling centres,	Heat-health command

	healthcare burden	hospital surge staffing, backup power	centre and dense-area water access.
Multan	High temperature and labour exposure	Worker rest breaks, hydration, heat alerts	Agricultural heat adaptation and outdoor labour policy.
Murree	Unusual heat anomaly risk	Tourism-season alerts, monitoring	Climate anomaly planning and emergency communication.

D. Additional Visual Validation

Additional notebook visuals were used to validate the modelling and interpretation workflow. The most important check is that the final leaderboard and actual-versus-predicted plots were regenerated after excluding target-derived variables. These visuals support the corrected conclusion that XGBoost and Random Forest outperform the Linear Regression baseline on the leakage-free feature matrix.

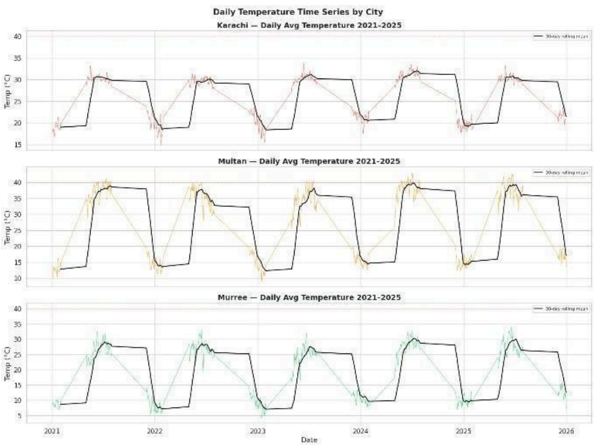

Fig. 15. Additional visual validation from the final notebook. The figure is retained as supporting evidence for the modelling section and should be read alongside the corrected leaderboard and actual-versus-predicted plots.

VII. KEY FINDINGS AND RECOMMENDATIONS

The modelling results support four actionable policy recommendations for city authorities and hospital planners:

- Heat-health alert systems should incorporate both temperature forecasts and load shedding schedules. The final feature importance results show that grid shortfall and outage hours are major leakage-free drivers, so emergency alerts should escalate when high heat and long outages occur together.
- Water distribution and cooling centre deployment should be pre-positioned 48–72 hours before a forecast crosses the 40°C threshold. The heatwave_streak feature confirms that multi-day heat accumulation is a stronger predictor than single-day peaks, meaning short-duration cooling interventions have limited protective effect.
- Karachi requires the highest overall preparedness priority because it has the highest composite vulnerability score and admissions burden. Multan still needs strong dry-heat and outdoor-labour protection because it has the highest average temperature profile.
- Hospital capacity planning should use the K-Means risk category of each day (Low, Medium, or High) as a triage signal. High-risk records should trigger excess-capacity protocols at least 24 hours in advance based on heat, outage, and admissions forecasts.

VIII. LIMITATIONS

Target leakage prevention: Earlier model experiments produced an unrealistically perfect Linear Regression result. The final notebook corrected this by excluding target-derived and post-outcome variables such as healthcare burden, labour productivity loss, economic loss, admissions-derived columns, predicted columns, and row identifiers from the admissions prediction matrix. These variables are retained only for EDA, impact analysis, vulnerability scoring, dashboards, and recommendations. Because admissions are monthly aggregates broadcast across daily rows, future work should still validate the model using independently collected daily hospital records.

News-scraping proxy: Heatstroke admission counts derived from Dawn and Express Tribune are subject to selection bias toward high-profile events. Official NDMA figures are preferred but known to undercount. Neither source provides a gold-standard ground truth.

Short observation window: The 2021–2025 window contains 2,310 daily observations across three cities. This is sufficient for model training, but too short to draw statistically significant long-term trend conclusions from rolling means alone. Extension to the full 1990–2024 NOAA historical archive would require additional data acquisition.

IX. FUTURE WORK

Future work should add verified hospital admissions at daily or hourly resolution, live weather APIs, satellite urban heat island indicators, GIS neighbourhood maps, power outage telemetry, and SMS/public-alert integration. The platform could also include role-based access for ministries, hospitals, utilities, and city planners.

A future version should include an automatic alerting engine that triggers when a city moves from Medium to High or Extreme risk. It should also log actions taken by departments, so the government can evaluate whether interventions reduced admissions.

LSTM architecture: The final notebook used a 14-day sequence window, but the LSTM performed poorly compared with tabular models. A future version should improve sequence forecasting using verified daily hospital admissions, longer history, exogenous climate forecasts, and careful hyperparameter tuning.

X. CONCLUSION

This study assembled, cleaned, and modelled a multi-source heatwave-health dataset for three Pakistani cities, advancing from 14 heterogeneous raw sources to a consolidated 2,310-row analytical dataset and ultimately to leakage-aware machine-learning models, clustering, dashboard exports, and the HeatShield Pakistan data product. Exploratory analysis confirmed major differences between Karachi, Multan, and Murree. The corrected modelling workflow showed that XGBoost and Random Forest outperformed the Linear Regression baseline after target leakage prevention. K-Means clustering produced operational Low, Medium, and High risk profiles that can support hospital and emergency management planning.

The findings confirm that heatstroke risk in Pakistan is not solely a temperature problem: infrastructure reliability and socioeconomic vulnerability are significant independent predictors. Policymakers should therefore design heat action plans that coordinate meteorological early-warning systems with grid management and targeted social protection rather than treating heatwave response as a purely meteorological function.

Future work should (a) apply piecewise or generalised additive regression to formally estimate heat-risk thresholds with confidence intervals; (b) extend the dataset to longer historical periods; (c) improve LSTM forecasting with longer verified daily admission sequences and exogenous climate variables; and (d) integrate real-time PMD forecasts and outage telemetry to move from retrospective analysis to operational early-warning deployment.

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